Problem 1.

思路部分.

设使用等距节点 $x_i = x_0 + ih$, $y_i = f(x_i)$ 作差分

向前差分: $\Delta y_i = y_{i+1} - y_i$

向后差分: $\nabla y_i = y_i - y_{i-1}$

中心差分: $\delta y_{i+\frac{1}{2}} = y_{i+1} - y_i$

引入移位算子 $E y_i \overset{def}{=} y_{i+1}$

$\Delta = E - 1$, $\nabla = 1 - E^{-1}$, $\delta = E^{\frac{1}{2}} - E^{-\frac{1}{2}}$

从而, n阶差分: $\Delta^n = (E-1)^n$, $\nabla^n = (1-E^{-1})^n$, $\delta^n = (E^{1/2} - E^{-1/2})^n$

向前差分

$i \backslash \Delta^i =$	1	E	E^2	E^3	E^4	E^5
1	-1	1				
2	1	-2	1			
3	-1	3	-3	1		
4	1	-4	6	-4	1	
5	-1	5	-10	10	-5	1

向后差分

$i \backslash \nabla^i =$	E^{-5}	E^{-4}	E^{-3}	E^{-2}	E^{-1}	0
1					-1	1
2				1	-2	1
3			-1	3	-3	1
4		1	-4	6	-4	1
5	-1	5	-10	10	-5	1

中心差分

$i \backslash \delta^i =$	$E^{-5/2}$	E^{-2}	$E^{-3/2}$	E^{-1}	$E^{-1/2}$	0	$E^{1/2}$	E^{1}	$E^{3/2}$	E^{2}	$E^{5/2}$
1					-1		1				
2				1		-2		1			
3			-1		3		-3		1		
4		1		-4		6		-4		1	
5	-1		5		-10		10		-5		1

用 Taylor 展开写出 $f(x+h) = \sum_{i=0}^{\infty} \frac{h^i}{i!} f^{(i)}(x) = e^{h\frac{d}{dx}} f(x) \overset{def}{=} e^{hD} f(x)$, $D \overset{def}{=} \frac{d}{dx}$

2阶

向前差分: $E^2 - 2E + 1 = (1 + 2hD + 2h^2D^2) - 2(1 + hD + \frac{h^2}{2}D^2) + 1) = h^2D^2$

$(E^2 - 2E + 1) f(x) = h^2 f''(x)$

$\therefore f''(x) = \frac{f(x+2h) - 2f(x+h) + f(x)}{h^2}$

向后, 中心差分. $f''(x) = \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}$

向后差分: $f''(x) = \frac{f(x) - 2f(x-h) + f(x-2h)}{h^2}$

因此所三种差分, 高阶的向前/向后差分均可如此推.

对于高阶的中心差分, 由于含 $\frac{1}{2}$, 所以不能直接写出.

令 $E^{(i+\frac{1}{2})} = \frac{1}{2}(E^i + E^{i+1})$

则: 三阶差分: $\delta^3 f(x) = -f(x-2h) + 2f(x-h) - 2f(x+h) + f(x+2h)$

五阶差分: $\delta^5 f(x) = -f(x-3h) + 4f(x-2h) - 5f(x-h) + 5f(x+h) - 4f(x+2h) + f(x+3h)$

以此类推即可

解答部分.

$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \frac{h^3}{6}f'''(x) + \frac{h^4}{24}f^{(4)}(x) + \frac{h^5}{120}f^{(5)}(x) + \frac{h^6}{720}f^{(6)}(x) + \frac{h^7}{5040}f^{(7)}(x) + o(h^7)$

$f(x-h) = f(x) - hf'(x) + \frac{h^2}{2}f''(x) - \frac{h^3}{6}f'''(x) + \frac{h^4}{24}f^{(4)}(x) - \frac{h^5}{120}f^{(5)}(x) + \frac{h^6}{720}f^{(6)}(x) - \frac{h^7}{5040}f^{(7)}(x) + o(h^7)$

$f(x+2h) = f(x) + 2hf'(x) + 2h^2f''(x) + \frac{4}{3}h^3f'''(x) + \frac{2}{3}h^4f^{(4)}(x) + \frac{4}{15}h^5f^{(5)}(x) + \frac{4}{45}h^6f^{(6)}(x) + \frac{8}{315}h^7f^{(7)}(x) + o(h^7)$

$f(x-2h) = f(x) - 2hf'(x) + 2h^2f''(x) - \frac{4}{3}h^3f'''(x) + \frac{2}{3}h^4f^{(4)}(x) - \frac{4}{15}h^5f^{(5)}(x) + \frac{4}{45}h^6f^{(6)}(x) - \frac{8}{315}h^7f^{(7)}(x) + o(h^7)$

$f(x+3h) = f(x) + 3hf'(x) + \frac{9}{2}h^2f''(x) + \frac{9}{2}h^3f'''(x) + \frac{27}{8}h^4f^{(4)}(x) + \frac{81}{40}h^5f^{(5)}(x) + \frac{81}{80}h^6f^{(6)}(x) + \frac{243}{560}h^7f^{(7)}(x) + o(h^7)$

$f(x-3h) = f(x) - 3hf'(x) + \frac{9}{2}h^2f''(x) - \frac{9}{2}h^3f'''(x) + \frac{27}{8}h^4f^{(4)}(x) - \frac{81}{40}h^5f^{(5)}(x) + \frac{81}{80}h^6f^{(6)}(x) - \frac{243}{560}h^7f^{(7)}(x) + o(h^7)$

$f(x+4h) = f(x) + 4hf'(x) + 8h^2f''(x) + \frac{32}{3}h^3f'''(x) + \frac{32}{3}h^4f^{(4)}(x) + \frac{128}{15}h^5f^{(5)}(x) + \frac{256}{45}h^6f^{(6)}(x) + o(h^6)$

$f(x-4h) = f(x) - 4hf'(x) + 8h^2f''(x) - \frac{32}{3}h^3f'''(x) + \frac{32}{3}h^4f^{(4)}(x) - \frac{128}{15}h^5f^{(5)}(x) + \frac{256}{45}h^6f^{(6)}(x) + o(h^6)$

$f(x+5h) = f(x) + 5hf'(x) + \frac{25}{2}h^2f''(x) + \frac{125}{6}h^3f'''(x) + \frac{625}{24}h^4f^{(4)}(x) + \frac{625}{24}h^5f^{(5)}(x) + \frac{3125}{144}h^6f^{(6)}(x) + o(h^6)$

$f(x-5h) = f(x) - 5hf'(x) + \frac{25}{2}h^2f''(x) - \frac{125}{6}h^3f'''(x) + \frac{625}{24}h^4f^{(4)}(x) - \frac{625}{24}h^5f^{(5)}(x) + \frac{3125}{144}h^6f^{(6)}(x) + o(h^6)$

二阶 向前: $f(x+2h) - 2f(x+h) + f(x)$

$= (f(x) + 2hf'(x) + 2h^2f''(x) + \frac{4}{3}h^3f'''(x) + \cdots) - 2(f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \frac{h^3}{6}f'''(x) + \cdots) + f(x)$

$= h^2f''(x) + h^3f'''(x) + o(h^3)$

$\therefore f''(x) = \frac{f(x+2h) - 2f(x+h) + f(x)}{h^2} + hf'''(x) + o(h)$

二阶 向后 $f(x) - 2f(x-h) + f(x-2h)$

$= (f(x) - 2hf'(x) + 2h^2f''(x) - \frac{4}{3}h^3f'''(x) + \cdots) - 2(f(x) - hf'(x) + \frac{h^2}{2}f''(x) - \frac{h^3}{6}f'''(x) + \cdots) + f(x)$

$= h^2f''(x) - h^3f'''(x) + o(h^3)$

$\therefore f''(x) = \frac{f(x) - 2f(x-h) + f(x-2h)}{h^2} - hf'''(x) + o(h)$

二阶 中心 $f(x+h) + f(x-h) - 2f(x)$

$= (f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \frac{h^3}{6}f'''(x) + \frac{h^4}{24}f^{(4)}(x)) + (f(x) - hf'(x) + \frac{h^2}{2}f''(x) - \frac{h^3}{6}f'''(x) + \frac{h^4}{24}f^{(4)}(x)) - 2f(x)$

$= h^2f''(x) + \frac{1}{12}h^4f^{(4)}(x)$

$\therefore f''(x) = \frac{1}{h^2}(f(x+h) - 2f(x) + f(x-h)) + \frac{1}{12}h^2f^{(4)}(x) + o(h^2)$

三阶向前 $f(x+3h) - 3f(x+2h) + 3f(x+h) - f(x)$

$= (f(x) + 3hf'(x) + \frac{9}{2}h^2f''(x) + \frac{9}{2}h^3f'''(x) + \frac{27}{8}h^4f^{(4)}(x))$

$-3(f(x) + 2hf'(x) + 2h^2f''(x) + \frac{4}{3}h^3f'''(x) + \frac{2}{3}h^4f^{(4)}(x))$

$+3(f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \frac{h^3}{6}f'''(x) + \frac{h^4}{24}f^{(4)}(x))$

$-f(x)$

$= h^3f'''(x) + \frac{3}{2}h^4f^{(4)}(x)$

$f'''(x) = \frac{1}{h^3}(f(x+3h) - 3f(x+2h) + 3f(x+h) - f(x)) + \frac{3}{2}hf^{(4)}(x) + o(h)$

三阶向后 $f(x) - 3f(x-h) + 3f(x-2h) - f(x-3h)$

$= f(x) - 3(f(x) - hf'(x) + \frac{h^2}{2}f''(x) - \frac{h^3}{6}f'''(x) + \frac{h^4}{24}f^{(4)}(x))$

$+3(f(x) - 2hf'(x) + 2h^2f''(x) - \frac{4}{3}h^3f'''(x) + \frac{2}{3}h^4f^{(4)}(x))$

$-(f(x) - 3hf'(x) + \frac{9}{2}h^2f''(x) - \frac{9}{2}h^3f'''(x) + \frac{27}{8}h^4f^{(4)}(x))$

$= h^3f'''(x) - \frac{3}{2}h^4f^{(4)}(x)$

$f'''(x) = \frac{1}{h^3}(f(x+3h) - 3f(x+2h) + 3f(x+h) - f(x)) - \frac{3}{2}hf^{(4)}(x) + o(h)$

三阶中心 $(f(x+2h) - f(x-2h)) - 2(f(x+h) - f(x-h))$

$= (4hf'(x) + \frac{8}{3}h^3f'''(x) + \frac{8}{15}h^5f^{(5)}(x)) - 2(2hf'(x) + \frac{1}{3}h^3f'''(x) + \frac{1}{60}f^{(5)}(x))$

$= 2h^3f'''(x) + \frac{1}{3}h^5f^{(5)}(x)$

$\therefore f'''(x) = \frac{1}{2h^3}(f(x+2h) - 2f(x+h) + 2f(x-h) - f(x-2h)) + \frac{1}{6}h^2f^{(5)}(x) + o(h^2)$

四阶向前 $f(x+4h) - 4f(x+3h) + 6f(x+2h) - 4f(x+h) + f(x)$

$= (f(x) + 4hf'(x) + 8h^2f''(x) + \frac{32}{3}h^3f'''(x) + \frac{32}{3}h^4f^{(4)}(x) + \frac{128}{15}h^5f^{(5)}(x))$

$-4(f(x) + 3hf'(x) + \frac{9}{2}h^2f''(x) + \frac{9}{2}h^3f'''(x) + \frac{27}{8}h^4f^{(4)}(x) + \frac{81}{40}h^5f^{(5)}(x))$

$+6(f(x) + 2hf'(x) + 2h^2f''(x) + \frac{4}{3}h^3f'''(x) + \frac{2}{3}h^4f^{(4)}(x) + \frac{4}{15}h^5f^{(5)}(x))$

$-4(f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \frac{h^3}{6}f'''(x) + \frac{h^4}{24}f^{(4)}(x) + \frac{h^5}{120}f^{(5)}(x)) + f(x)$

$= h^4f^{(4)}(x) + 2h^5f^{(5)}(x)$

$\therefore f^{(4)}(x) = \frac{1}{h^4}(f(x+4h) - 4f(x+3h) + 6f(x+2h) - 4f(x+h) + f(x)) + 2hf^{(5)}(x) + o(h)$

四阶导数 $f(x-4h)-4f(x-3h)+6f(x-2h)-4f(x-h)+f(x)$

$$=\left(f(x)-4hf'(x)+8h^2f''(x)-\frac{32}{3}h^3f'''(x)+\frac{64}{3}h^4f^{(4)}(x)-\frac{128}{15}h^5f^{(5)}(x)\right)$$

$$-4\left(f(x)-3hf'(x)+\frac{9}{2}h^2f''(x)-\frac{9}{2}h^3f'''(x)+\frac{27}{8}h^4f^{(4)}(x)-\frac{81}{80}h^5f^{(5)}(x)\right)$$

$$+6\left(f(x)-2hf'(x)+2h^2f''(x)-\frac{4}{3}h^3f'''(x)+\frac{2}{3}h^4f^{(4)}(x)-\frac{4}{15}h^5f^{(5)}(x)\right)$$

$$-4\left(f(x)-hf'(x)+\frac{h^2}{2}f''(x)-\frac{h^3}{6}f'''(x)+\frac{h^4}{24}f^{(4)}(x)-\frac{h^5}{120}f^{(5)}(x)\right)+f(x)$$

$$=h^4f^{(4)}(x)-2h^5f^{(5)}(x)$$

$$\therefore f^{(4)}(x)=\frac{1}{h^4}\left(f(x-4h)-4f(x-3h)+6f(x-2h)+4f(x-h)+f(x)\right)-2hf^{(5)}(x)+o(h)$$

四阶中心 $f(x+2h)-4f(x+h)+6f(x)-4f(x-h)+f(x-2h)$

$$=\left(2f(x)+4hf''(x)+\frac{4}{3}h^3f^{(4)}(x)+\frac{8}{45}h^5f^{(6)}(x)\right)$$

$$-4\left(2f(x)+hf''(x)+\frac{1}{12}h^4f^{(4)}(x)+\frac{1}{360}h^5f^{(6)}(x)\right)+6f(x)$$

$$=h^4f^{(4)}(x)+\frac{1}{6}h^6f^{(6)}(x)$$

$$\therefore f^{(4)}(x)=\frac{1}{h^4}\left(f(x+2h)-4f(x+h)+6f(x)-4f(x-h)+f(x-2h)\right)+\frac{1}{6}h^2f^{(6)}(x)+o(h^2)$$

五阶向前 $f(x+5h)-5f(x+4h)+10f(x+3h)-10f(x+2h)+5f(x+h)-f(x)$

$$=\left(f(x)+5hf'(x)+\frac{25}{2}h^2f''(x)+\frac{125}{6}h^3f'''(x)+\frac{625}{24}h^4f^{(4)}(x)+\frac{3125}{120}h^5f^{(5)}(x)+\frac{3125}{720}h^6f^{(6)}(x)\right)$$

$$-5\left(f(x)+4hf'(x)+8h^2f''(x)+\frac{32}{3}h^3f'''(x)+\frac{64}{3}h^4f^{(4)}(x)+\frac{128}{15}h^5f^{(5)}(x)+\frac{256}{45}h^6f^{(6)}(x)\right)$$

$$+10\left(f(x)+3hf'(x)+\frac{9}{2}h^2f''(x)+\frac{9}{2}h^3f'''(x)+\frac{27}{8}h^4f^{(4)}(x)+\frac{81}{80}h^5f^{(5)}(x)+\frac{81}{80}h^6f^{(6)}(x)\right)$$

$$-10\left(f(x)+2hf'(x)+2h^2f''(x)+\frac{4}{3}h^3f'''(x)+\frac{2}{3}h^4f^{(4)}(x)+\frac{4}{15}h^5f^{(5)}(x)+\frac{4}{45}h^6f^{(6)}(x)\right)$$

$$+5\left(f(x)+hf'(x)+\frac{h^2}{2}f''(x)+\frac{h^3}{6}f'''(x)+\frac{h^4}{24}f^{(4)}(x)+\frac{h^5}{120}f^{(5)}(x)+\frac{h^6}{720}f^{(6)}(x)\right)$$

$$-f(x)$$

$$=h^5f^{(5)}(x)+\frac{5}{2}h^6f^{(6)}(x)$$

$$\therefore f^{(5)}(x)=\frac{1}{h^5}\left(f(x+5h)-5f(x+4h)+10f(x+3h)-10f(x+2h)+5f(x+h)-f(x)\right)+\frac{5}{2}hf^{(6)}(x)+o(h)$$

五阶后向 $-f(x-5h)+5f(x-4h)-10f(x-3h)+10f(x-2h)-5f(x-h)+f(x)$

$$=-\left(f(x)-5hf'(x)+\frac{25}{2}h^2f''(x)-\frac{125}{6}h^3f'''(x)+\frac{625}{24}h^4f^{(4)}(x)-\frac{3125}{120}h^5f^{(5)}(x)+\frac{3125}{720}h^6f^{(6)}(x)\right)$$

$$+5\left(f(x)-4hf'(x)+8h^2f''(x)-\frac{32}{3}h^3f'''(x)+\frac{64}{3}h^4f^{(4)}(x)-\frac{128}{15}h^5f^{(5)}(x)+\frac{256}{45}h^6f^{(6)}(x)\right)$$

$$-10\left(f(x)-3hf'(x)+\frac{9}{2}h^2f''(x)-\frac{9}{2}h^3f'''(x)+\frac{27}{8}h^4f^{(4)}(x)-\frac{81}{80}h^5f^{(5)}(x)+\frac{81}{80}h^6f^{(6)}(x)\right)$$

$$+10\left(f(x)-2hf'(x)+2h^2f''(x)-\frac{4}{3}h^3f'''(x)+\frac{2}{3}h^4f^{(4)}(x)-\frac{4}{15}h^5f^{(5)}(x)+\frac{4}{45}h^6f^{(6)}(x)\right)$$

$$-5\left(f(x)-hf'(x)+\frac{h^2}{2}f''(x)-\frac{h^3}{6}f'''(x)+\frac{h^4}{24}f^{(4)}(x)-\frac{h^5}{120}f^{(5)}(x)+\frac{h^6}{720}f^{(6)}(x)\right)$$

$$+f(x)$$

$$=h^5f^{(5)}(x)-\frac{5}{2}h^6f^{(6)}(x)$$

$$\therefore f^{(5)}(x)=\frac{1}{h^5}\left(f(x)-5f(x-h)+10f(x-2h)-10f(x-3h)+5f(x-4h)-f(x-5h)\right)-\frac{5}{2}hf^{(6)}(x)+o(h)$$

五阶中心 $\left(f(x+3h)-f(x-3h)\right)-4\left(f(x+2h)-f(x-2h)\right)+5\left(f(x+h)-f(x-h)\right)$

$$=\left(6hf'(x)+9h^3f'''(x)+\frac{81}{20}h^5f^{(5)}(x)+\frac{243}{280}h^7f^{(7)}(x)\right)$$

$$-4\left(4hf'(x)+\frac{8}{3}h^3f'''(x)+\frac{8}{15}h^5f^{(5)}(x)+\frac{16}{315}h^7f^{(7)}(x)\right)$$

$$+5\left(2hf'(x)+\frac{1}{3}h^3f'''(x)+\frac{1}{60}h^5f^{(5)}(x)+\frac{1}{1890}h^7f^{(7)}(x)\right)$$

$$=2h^5f^{(5)}(x)+\frac{1}{5}h^7f^{(7)}(x)$$

$$\therefore f^{(5)}(x)=\frac{1}{2h^5}\left(f(x+3h)-4f(x+2h)+5f(x+h)-5f(x-h)+4f(x-2h)-f(x-3h)\right)+\frac{1}{5}h^2f^{(7)}(x)+o(h^2)$$